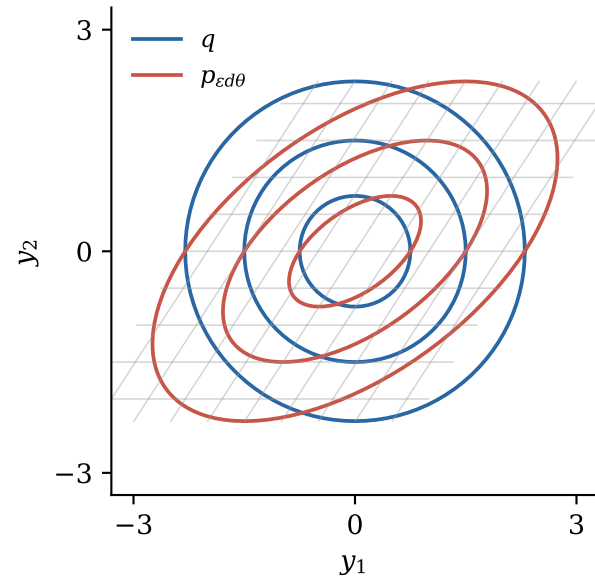


Shear: the prescribed motion and the motion WO needs

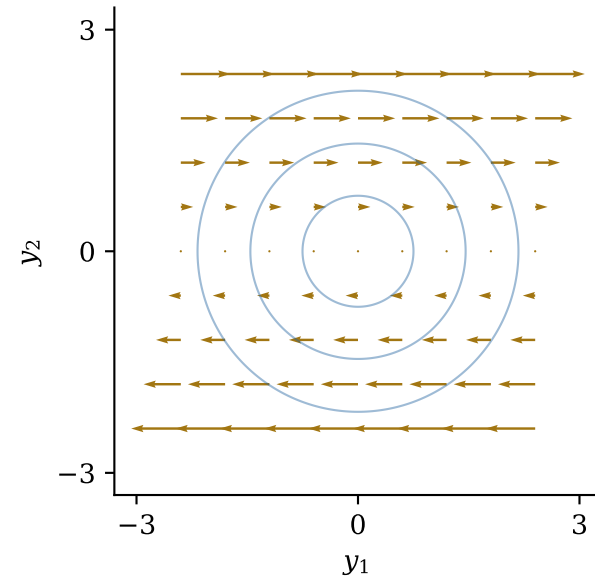
$$\phi_\theta(x) = (x_1 + \theta x_2, x_2), \quad \theta = 0, \quad d\theta = 1$$

(a) A finite shear



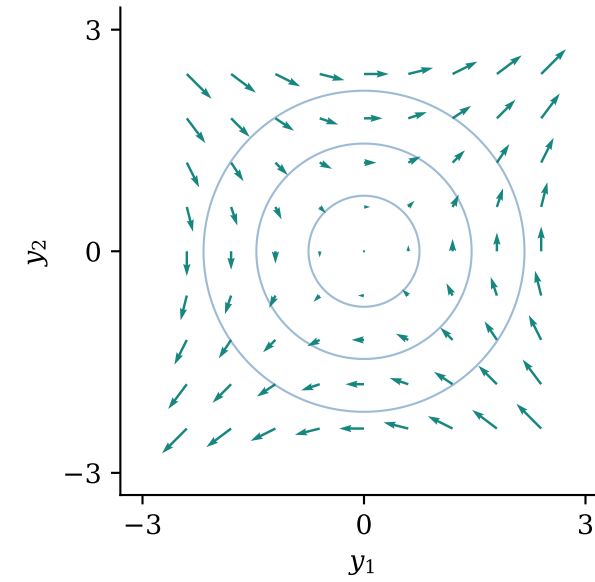
$\epsilon d\theta = 0.65$ (display only)

(b) Induced velocity



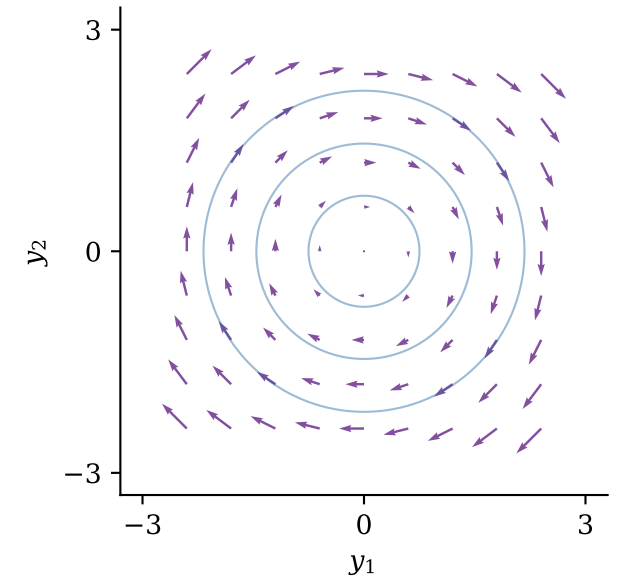
$v = (y_2, 0)$

(c) Gradient projection



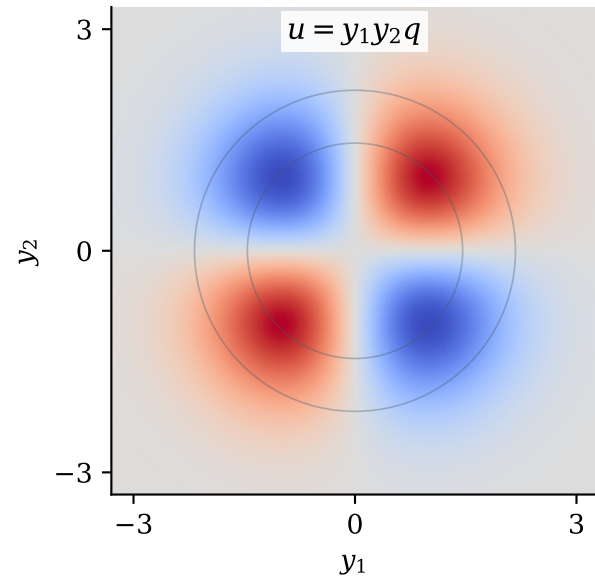
$\nabla \psi = \frac{1}{2}(y_2, y_1)$

(d) Density-preserving rotation

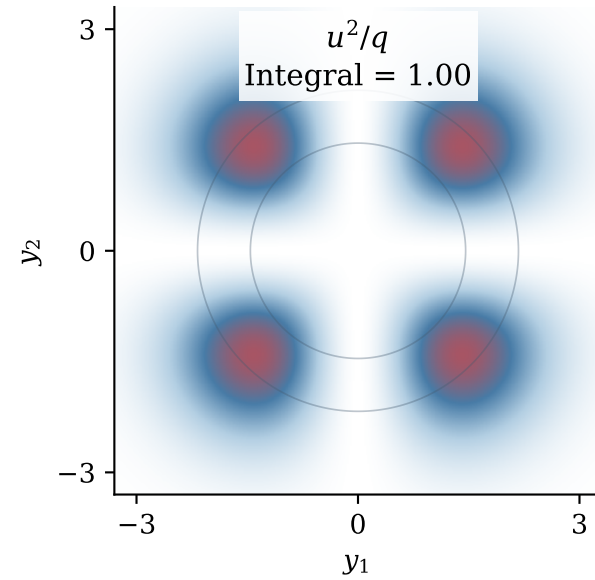


$w = \frac{1}{2}(y_2, -y_1)$

(e) Density variation

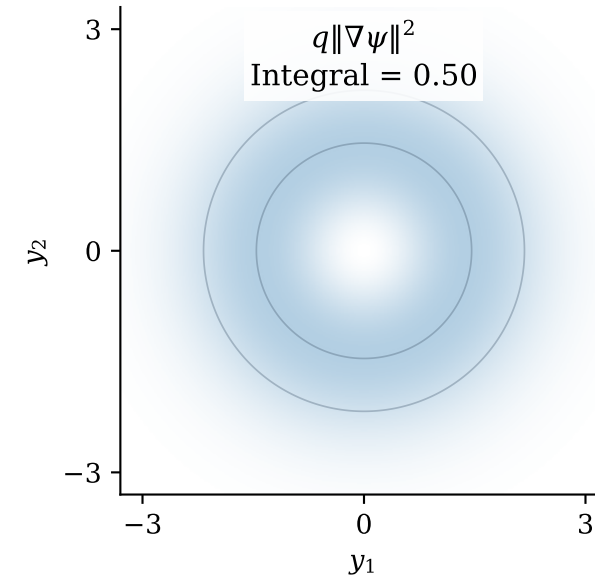


(f) Fisher-Rao



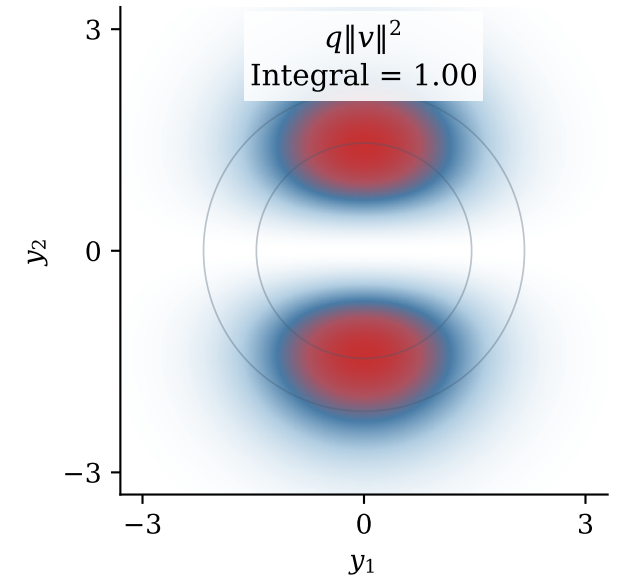
u^2/q
Integral = 1.00

(g) Wasserstein-Otto



$q \|\nabla \psi\|^2$
Integral = 0.50

(h) Prescribed-motion energy



$q \|v\|^2$
Integral = 1.00

